

MATH9102: Fundamentals of Data Analysis

Week 1: Fundamentals

The Process, the Variables, and Describing Data

Today

1. **Probability, Statistics, Data Science** — three people, one dice table
2. **The Process** — from an observation to a reported finding
3. **Variables and Measurement** — the decision everything else rests on
4. **Population and Sample**
5. **Describing Data** — centre, spread, shape
6. **Getting Started in Python**
7. **Misleading Statistics** — and how not to

The point of the week

Nothing here is hard arithmetic. All of it is judgement, and the judgement is the part that has to be defended.

What This Module Is About

Analysing data: evaluating whether it can answer your question, and saying what you found in a way somebody can act on.

You will learn to

- Design an analysis around a question
- Judge whether data can support it
- Describe data so its limits are visible
- Run and interpret the standard tests
- Build and criticise simple models
- Report findings for a reader who is not you

What it is not

This is not a module in learning Python, and not a module in visualisation. Python is the tool; the subject is the analysis.

You will be given working code. What matters is what you do with it.

How You Are Assessed

100% continuous assessment. There is no examination.

Part	Weight	What it asks for	Due
Phase 1	40%	Evaluate a dataset's suitability for multivariate analysis; explore and summarise it; justify your methodological choices	Week 8
Phase 2	60%	Design and implement a data science workflow; apply multivariate techniques; evaluate the workflow critically	Week 14

Read the verbs

Evaluate, justify, critically assess, communicate. None of them is *compute*. Running the test is the easy part.

Probability, Statistics, Data Science

Three People at a Dice Table

A six-sided die. Three specialists watch the same game.

The probabilist

- Assumes a fair die
- Each face has probability $1/6$
- Computes the chance of any outcome from theory

The statistician

- Asks whether the die *is* fair
- Records many rolls
- Tests the observed frequencies against what fairness predicts

The data scientist

- Starts from data, not assumptions
- Explores it, estimates the probabilities, simulates
- Reports an estimate *with* its uncertainty

The difference in one sentence each

"The chance is $1/6$." "There is no evidence the die is unfair." "The estimated chance of a six is 0.19, with a 95% interval from 0.12 to 0.26."

They are not rivals. They are a pipeline.

- **Probability** gives the framework — what “fair” would look like
- **Statistics** is the bridge — does reality match the framework?
- **Data science** is the application — explore at scale, find patterns, act

Where this module sits

Mostly in the middle, with one foot on either side. You need enough probability to know what a test assumes, and enough of the applied side to get an answer out of a real dataset.

Watch out

The data scientist's answer is the most useful *and* the easiest to overstate. An estimate without its uncertainty is not a finding.

What Statistics Buys You

A way to compare things objectively, and a way to say how sure you are.

Two groups of numbers:

6, 1, 8, 3, 5, 4, 9

8, 3, 4, 2, 7, 1, 4

Which is larger? Eyeballing them gets you an impression. Averaging them gets you a number two people can agree on.

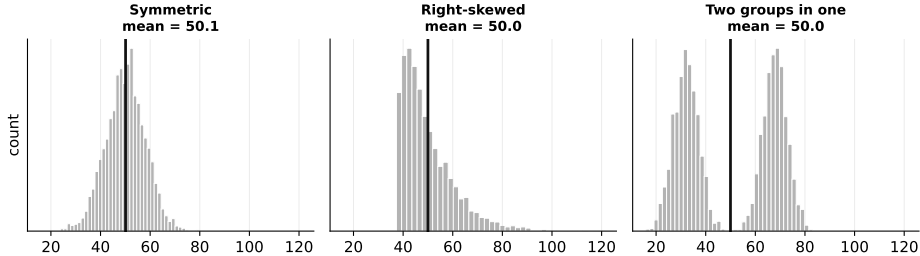
Two things you get

Summary — reduce many numbers to a few, so a reader does not have to look at every case.

Inference — say something about a population from a sample, and quantify how much to trust it.

And the cost: **every summary throws information away**. The skill is knowing what you just discarded.

What an Average Hides



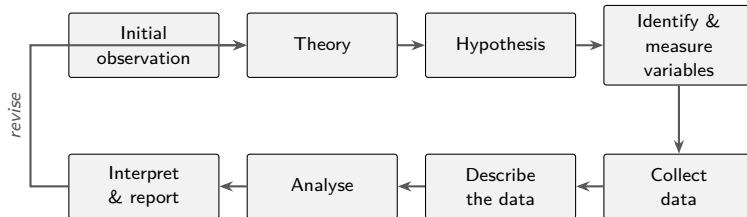
Three distributions. One mean. A delivery service averaging one day might deliver most parcels in hours and a few in a week.

Watch out

There is an old story about the statistician who drowned crossing a lake with an average depth of three feet.

The Process

The Process, End to End



It starts with **something that needs explaining**
— something you noticed, experienced, or read.
Not with a dataset you happen to have.

The loop is the point

What you find feeds back into the theory. A finding that revises nothing was probably not a question.

The system of concepts, assumptions and expectations you bring to the problem — whether or not you have written it down.

It is a tentative model of what is out there and what is going on with it. It decides:

- What counts as a relevant variable
- What data you need to collect
- What you will treat as a plausible explanation

Why write it down

Because you have one either way.

Unwritten, it is a set of assumptions you cannot examine, and neither can anyone reviewing your work.

Why the Framework Matters

Bernd Heinrich spent a summer studying ant lions — insects that trap ants in pits they dig.

- His results differed sharply from everyone else's
- Revolutionary? Or wrong?
- He repeated the work the following summer
- His team had misunderstood the animals' behaviour — specifically the time frame over which it happens

The lesson

“Even carefully collected results can be misleading if the underlying context of assumptions is wrong.”

The data were fine. The analysis was fine. The framework was wrong, and nothing downstream could detect that.

Maxwell, J. A. (2013). *Qualitative research design: an interactive approach*, 3rd ed., pp. 39–72.

Theory and Hypothesis

Theory

- A general principle that explains known findings, and from which new predictions can be generated
- *Computer science attracts students with strong mathematical ability*

Hypothesis

- A specific prediction from that theory
- States a relationship between concepts you can actually observe
- *Applicants to the MSc will have higher mathematical ability than the general population*

The test of a good hypothesis

You can say, *in advance*, what result would count against it. If no imaginable data would embarrass it, it is not a hypothesis.

Identifying the Variables

A variable is anything that can vary — anything you can observe, change, manipulate, calculate or simulate.

Hypothesis

The number of hours a student works for an employer affects the number of hours they spend on independent study.

Variables

- Hours worked for an employer
- Hours of independent study

Do this before you touch data

Naming the variables is how you find out whether your hypothesis is testable at all. A concept you cannot turn into something measurable is not yet a hypothesis.

Predictor and Outcome

Consumption of Coca-Cola improves a student's ability to concentrate during lectures.

Coca-Cola consumption

- The proposed **cause**
- **Predictor** variable
- Manipulated, in an experiment
- Also called the *independent* variable

Ability to concentrate

- The proposed **effect**
- **Outcome** variable
- Measured, not manipulated
- Also called the *dependent* variable

Watch out

“Independent” and “dependent” are historical names from experimental design. In observational data nothing is manipulated and nothing is independent — prefer **predictor** and **outcome**, which claim less.

Confounding Variables

A variable that influences others in your study, whether or not you measured it.

Two variables are confounded when the effect of one cannot be separated from the effect of the other.

Example. Petrol used and time taken to commute. Both rise with congestion, which you probably did not record.

Back to the Coca-Cola

Dosage and timing you can control.
Caffeine tolerance, sleep the night before, how interesting the lecture is — you cannot, and each of them moves the outcome.

The honest position

We will not be able to demonstrate cause and effect. We will be able to establish association, describe it, and say what would have to be true for a causal reading to hold.

Measurement Error, Validity, Reliability

Measurement error

- **Systematic:** the scale reads 0.1 kg heavy for everyone
- **Random:** the scale is accurate, but athletes are variably hydrated and dressed

Validity

- Does the instrument measure what it claims to?
- *Content:* does it cover the whole construct?
- *Ecological:* does it hold outside the study?

Reliability

- Does it give the same answer under the same conditions?
- *Test-retest:* the same people, twice, same result

They fail differently

A *random* error makes your estimates noisier. A *systematic* one makes them wrong, and no sample size fixes it. An instrument can be perfectly reliable and measure the wrong thing.

Variables and Measurement

Variables, Cases and Datasets

The vocabulary

- A **variable** holds information about one concept: income, education level, whether someone smokes
- A **case** (or observation) is one thing you measured
- A **dataset** is cases down the rows, variables across the columns

Quantitative or qualitative

- **Quantitative** — a measurement. Arithmetic is meaningful
- **Qualitative** (categorical) — a description. Arithmetic is not, *even when the categories are stored as numbers*

An example dataset

Forty MSc students, each with an age, a previous education level and a confidence rating.
Forty cases; three variables; one measurement per case per variable.

Nominal

Categories that name, with no order among them.

Country of origin; type of residence; eye colour.

Categories must be:

- **mutually exclusive** — every case fits one, and only one
- **collectively exhaustive** — every case fits at least one

Watch out

You survey religious belief with categories for Catholic, Protestant, Jewish, Muslim, Hindu and Other.

A respondent is an atheist. Where do they go? “Other” is exhaustive, but it now means two different things, and you can never separate them again.

Categories that name *and* rank — but with unequal, unknown gaps.

- Finishing position: 1st, 2nd, 3rd
- Strongly disagree → strongly agree
- Highest level of education completed

Someone who finished second level has less education than someone with a degree. **How much less is not defined.**

What follows

You may rank, take a median, and report frequencies.

You may *not* average — and software will let you, because the categories are stored as 1 to 5 and it cannot tell the difference.

Discrete and Continuous

Discrete

- Isolated points on a number line
- Usually **counted**
- Students in the room; the face of a die; children in a household

Continuous

- Any value in a range
- Usually **measured**
- Height; time spent concentrating; rate of flow through a pipe

A key note

Your dataset does not have to contain every possible value of a variable. It does have to be *possible* for a value to appear if it occurs in the world you are modelling. A household with six children is rare; an instrument that stops counting at four has put a hole in the design, and the absence of sixes in your data is a fact about the instrument rather than about households.

Interval and Ratio

Interval

- Ordered, with **equal and fixed** gaps between points
- **No true zero** — zero is a point on the scale, not an absence
- Add and subtract, yes. Multiply and divide, no

Wake-up time to the nearest quarter hour. 9:00 to 9:15 is the same gap as 18:00 to 18:15. But 0:00 is not the absence of time.

In practice

Most tests that want “interval” are happy with either. The distinction matters when you want to say *twice as much*.

Ratio

- Everything interval has, plus
- A **true zero** meaning none of the thing
- So ratios are meaningful: one value can be twice another

Money spent on lunch. Spending nothing is spending nothing, and 14.02 euro really is twice 7.01.

The Level of Measurement Is Your Judgement

No file format records it. No function returns it. You decide, and you defend the decision.

Level	Legitimate summaries	What it rules out
Nominal	counts, percentages, mode	any arithmetic on the codes
Ordinal	the above, plus median and range	means, standard deviations
Interval / ratio	the above, plus mean, SD, variance	(ratio statements need a true zero)

Watch out

Python will compute the mean of anything numeric, including a category code. The answer is arithmetic; it is not information.

Population and Sample

Population and Sample

Population

- Every individual or object of interest — the world you are modelling
- Finite or infinite
- *All registered voters in Ireland*

Sample

- A part of the population, standing in for it
- *Two thousand of those voters*

Why you almost always have a sample

The population is large — Ireland has millions of registered voters — and it will not hold still. People turn eighteen and are added; people die and are removed. By the time you finished counting it, it would be a different population.

Population or Sample? Two Judgements

Employee salaries

You have every employee record.

- Analysing salaries *on one day*: you have the population
- Analysing salaries *over a year*: people join, leave, are promoted — you have a sample of a moving target

Customer satisfaction

You want every customer from the past year.

- That set is the population
- Could you contact all of them? Unlikely
- Would they all reply? Certainly not
- So: a sample, and you must argue it is representative

The general rule

Whether you have a population or a sample is a fact about **your question**, not about how many rows you happen to have.

What Makes a Sample Good

It has to be a fair proxy. If it is not, it is biased — and bias is not fixed by collecting more of it.

Simple random sample

Every possible sample of a given size has an equal chance of being the one you get.

- Every value that occurs in the population gets its fair chance to appear
- The researcher's preferences cannot influence selection

A biased sampling method

One that produces data differing *systematically* from the population it came from.

Note that even a genuinely random sample may fail to reflect the diversity of a population — randomness protects against systematic bias, not against bad luck.

How Big Should the Sample Be?

Too small

- Conclusions that do not hold for the population
- Real effects you cannot detect

Too large

- Wasteful, if a smaller sample would have done
- Sometimes an ethical problem, depending on what you are doing to the participants

Not a question you answer by feel

Sample size follows from how big an effect you want to detect and how sure you want to be. We return to it once we can say what those words mean.

Watch out

A large sample does not rescue a bad one. Ten thousand responses from people who chose to answer still describe people who chose to answer.

Describing Data

Describe Before You Analyse

You must describe your data before you analyse it, and your reader must see the description.

Why

- Your reader can see what your conclusions rest on
- They can see what *constrains* those conclusions
- They should not have to look at every case to follow you

How

- Start with a **picture**
- Add numbers that summarise the centre *and* the variability
- Say what is missing

Two words worth separating

Descriptive statistics summarise the data you have. **Inferential** statistics reason from a sample to a population. This week is entirely the first; from week 3 it is mostly the second.

Central Tendency: Three Answers

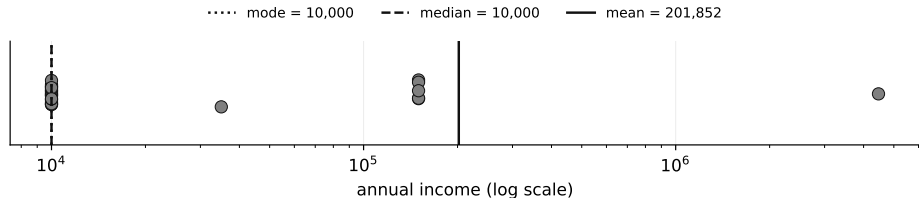
A single value that the others tend to cluster around — and three different ways to pick it.

	What it is	When to reach for it
Mode	the most frequent value	nominal data — it is the only one available
Median	the middle value once sorted	ordinal data, or skewed continuous data
Mean	the sum divided by the count	symmetric continuous data

The trade-off in one line

The mean uses every observation, which is its strength when the distribution is well behaved and its weakness when it is not. The median ignores how extreme the extremes are, which is the same property seen from the other side.

A Neighbourhood

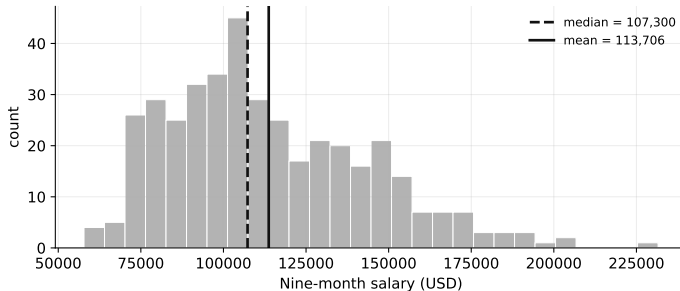


Twenty people on a modest wage, one on rather more, five professionals, and a film star. Every dot is a person.

Read where the mean falls

Nobody on that street earns anything near it. The arithmetic is correct; the summary is a lie by omission.

The Worked Dataset



Nine-month academic salaries at one US college.
A long tail to the right, so the **mean sits above the median**.

A rule of thumb

Mean noticeably above median means a right tail. Below means a left tail. Look at the picture before you trust either number.

Dispersion: How Spread Out?

A centre without a spread is half a description.

Range

- Largest minus smallest
- Uses two observations and ignores every other one

Interquartile range

- The width of the middle half
- Ignores the tails by construction — which is exactly why it survives them

Variance

- Average *squared* distance from the mean
- Squared because raw deviations cancel to zero — and squares rather than absolute values, because the rest of statistics is built on them

Watch out

Variance is in *squared* units. The variance of a salary is in dollars squared, which nobody has an intuition for.

Standard Deviation

The square root of the variance — back in the units of the original variable, which is why it is the one you report.

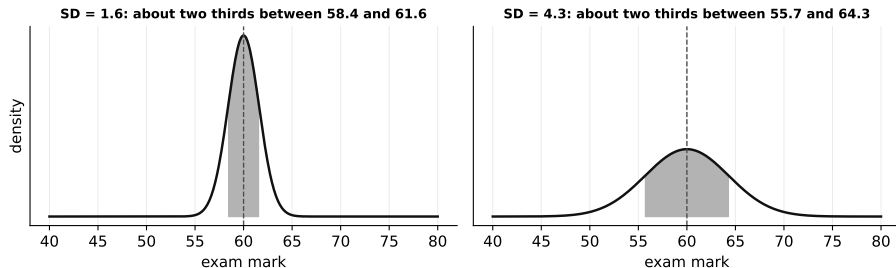
- How far a typical case sits from the mean
- Larger means more spread out
- For a roughly *normal* distribution: about two thirds of cases within one SD of the mean, almost all within three — week 2 makes this exact

One divisor, two defaults

Divide by one *less* than the number of observations: estimating the mean used up one piece of information.

pandas does this by default. numpy does not. They will silently disagree.

Same Mean, Different Story

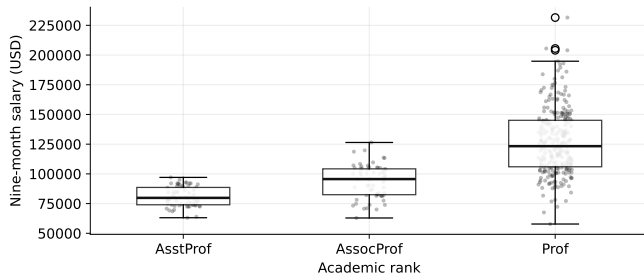


Two programmes, the same number of students, the same mean exam mark. Did students on both perform equally well?

No

One programme has almost everybody near the average. The other has students failing and students excelling. The mean cannot tell them apart.

Describing Within Groups



A continuous variable described within the levels of a categorical one. Read the group sizes, the spread inside each group, and the overlap.

Watch out

This is a description, not a test. We are not yet entitled to say the groups *differ* — only what these data look like.

Getting Started in Python

The Tools

Python	the language
pandas	tabular data — a spreadsheet you can program
scipy, statsmodels, pingouin	the statistics
matplotlib	the figures
math9102	this module's own package

Why the module package is thin

It supplies the datasets, the house figure style and the reporting functions — the things you would otherwise copy and paste every week.

Everything statistical uses the standard libraries directly, because those are what you will still be using after this module ends.

Why the Environment Is Pinned

Every version of every package is fixed in requirements.lock.

An analysis that gives one answer on your machine and another on mine is not an analysis. **The version of a package is part of the method**, and belongs in the write-up with everything else.

Watch out

You will meet code that installs packages at run time inside a wrapper that suppresses all errors. When the install fails, the script continues and breaks later, somewhere unrelated, for no visible reason.

Values, Series and DataFrames

```
# A Series holds many values of one kind. Arithmetic applies to all of them.  
age = pd.Series([28, 48, 47, 71, 22, 80, 48, 30, 31])  
purchase = pd.Series([20, 59, 2, 12, 22, 160, 34, 34, 29])  
  
# A DataFrame holds several Series side by side: one row per case.  
techsales = pd.DataFrame({"age": age, "purchase": purchase})  
  
# Select rows by a condition, not by position.  
techsales[techsales.age > 40]
```

- A **Series** is one variable: many values of one kind
- A **DataFrame** is many Series sharing an index — one row per case

Note what is missing

No loop. You describe the rows you want and pandas applies the operation to all of them.

Load Data by Name, Never by Path

```
# Datasets are loaded by name, never by path. A path is a fact about one
# computer; a name is a fact about the analysis.
salaries = m9.load_salaries()

print(salaries.shape)
salaries.dtypes

# dtypes tell you how Python stores the column. They do NOT tell you the level
# of measurement - that is your judgement, and it decides what is legitimate.
salaries.head()
```

This is a rule, not a convenience

An absolute path is a fact about one computer: whose account, which drive, which folder. Hardcode one and the analysis runs on that machine and nowhere else. A dataset name is a fact about the analysis, and it travels.

Summarising a Categorical Variable

```
# For a categorical variable there is one honest summary: how many, what share.  
m9.frequency(salaries, "rank")  
  
# Two categoricals give a contingency table.  
m9.crosstab(salaries, "discipline", "rank")  
  
# Percentages within rows: of the people in this discipline, what share are at  
# each rank? Column-wise answers a different question. Always say which.  
m9.crosstab(salaries, "discipline", "rank", normalize="index")
```

Counts are hard to compare when the row totals differ. Percentages fix that — and immediately raise the question of *which* total.

Watch out

A table of percentages that does not say which way it was computed is unreadable. Row and column versions of one table tell different stories.

```
# Three answers to "the typical value", and they disagree when data are skewed.  
salary = salaries.salary  
  
print(f"mean = {salary.mean():,.0f}")  
print(f"median = {salary.median():,.0f}")  
print(f"mode = {salary.mode().iloc[0]:,.0f}")  
  
# Mean above median is the signature of a tail to the right.
```

What to look for

Run all three and compare them. The gap between the mean and the median is the cheapest skewness diagnostic there is, and it costs you one line.

Spread, and a Trap

```
# A centre means little without a spread.  
print(f"range = {salary.max() - salary.min():,.0f}")  
print(f"IQR = {salary.quantile(.75) - salary.quantile(.25):,.0f}")  
print(f"SD = {salary.std(ddof=1):,.0f}")  
  
# ddof is the divisor correction. pandas defaults to 1, numpy defaults to 0,  
# and they will silently disagree. Know which you are using.  
print(f"pandas std = {salary.std():.3f}")  
print(f"numpy std = {np.std(salary):.3f}")
```

ddof

pandas defaults to ddof=1 and numpy to ddof=0. On a large dataset the difference is invisible; on a small one it is not. Two lines of your own code can disagree with each other.

Everything At Once

```
# Everything at once, for the variables you name.  
m9.describe(salaries, ["salary", "yrs_since_phd", "yrs_service"])  
  
# skew and kurtosis describe SHAPE. They are descriptions, not verdicts.  
m9.describe_by(salaries, "salary", "discipline")
```

Skew — positive is a tail to the right, negative to the left, zero symmetric.

Kurtosis — how much of the variance comes from rare extreme values.

Watch out

Both are *descriptions of shape*, not pass/fail tests. Week 3 returns to what they can and cannot decide.

Missing Data, on Day One

```
# What a complete-case analysis would cost, for the variables you are using -  
# never for the whole dataframe.  
cheap = m9.missingness(msleep, ["sleep_total", "vore"])  
print(cheap.attrs["summary"])  
  
expensive = m9.missingness(msleep, ["sleep_cycle", "brainwt", "vore"])  
print(expensive.attrs["summary"])  
  
# Same data, same function. Choose the variables first, then count the cost.
```

Choose the variables, then count the cost

The same dataset costs almost nothing or almost everything depending on which columns you name. Dropping every incomplete row *before* choosing your variables throws away cases for columns your analysis never touches.

Misleading Statistics

Statistics Can Mislead

The misuse of numerical data — sometimes deliberate, more often not.

Collection

- Who was asked
- What they were asked
- Who did not reply

Processing

- Which cases were dropped
- Which comparisons were tried
- Which were reported

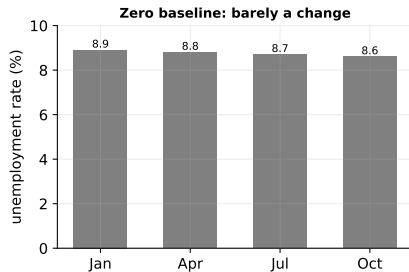
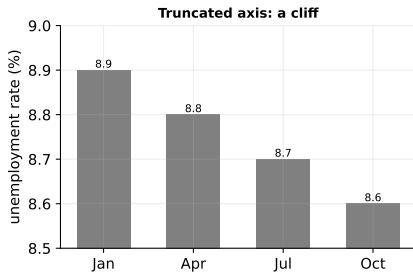
Presentation

- What the axes do
- What the picture implies
- What the caption does not say

The useful habit

For any statistic you meet: **who was measured, what were they asked, and what is the denominator?** Most misleading statistics fail one of those three before you need any mathematics at all.

The Truncated Axis



The same four numbers, twice. On the left the change looks catastrophic; on the right it is a rounding error.

Watch out

Neither chart is false. Read the axis before you read the bars — and when you draw one, ask what your choice of axis is doing to the reader.

Percentages Without a Denominator

“More than 80% of dentists recommend Colgate”

- Surveyed dentists could recommend *several* brands
- So every competing brand could score the same
- The UK Advertising Standards Authority ruled against it, 2007

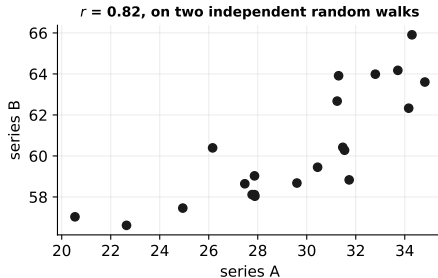
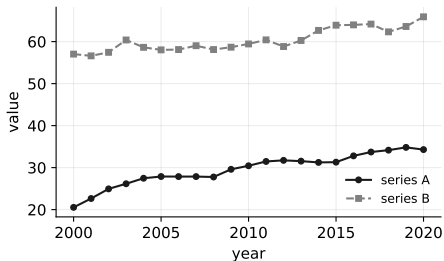
Percentages that sum to 485%

- Respondents could pick more than one option
- Perfectly legitimate — and it makes a pie chart meaningless, because the slices are not parts of a whole

And the subtler version

A chart shows the ethnic breakdown of undergraduates and invites you to conclude one group is under-represented. Under-represented *relative to what*? Without the breakdown of the eighteen-to-twenty-four-year-old population, the numbers support no conclusion in either direction.

Correlation Without Connection



Two independent random walks. Neither has any connection to the other. The correlation is real arithmetic about a relationship that does not exist.

Why it happens

Anything that trends over time correlates with anything else that trends over time.

How Not To Fall In

The common failures

- Faulty survey design — who, what, when, how
- Insufficient or unrepresentative samples
- Correlations that mean nothing in the real world
- Selective reporting of one period or subgroup
- **Data dredging** — hunting for relationships with no hypothesis

The defences

- **Know your question before you start.** No fishing
- Understand the conceptual framework of work you rely on
- Report what you did, including what did not work
- Look for the source and context of any statistic you quote

Worth knowing

A meta-analysis of survey data found that about a third of scientists admitted to some questionable research practice — modifying results, interpreting data selectively, or dropping observations on a hunch. Mostly not fraud; mostly the absence of a rule that would have stopped them.

Fanelli, D. (2009). How many scientists fabricate and falsify research? *PLoS ONE* 4(5): e5738.

Close

What you should take from this week

You **SHOULD** be able to

- State a question, a theory and a hypothesis, and name the variables they imply
- Give any variable a level of measurement, and defend it
- Say whether you have a population or a sample, and why it matters
- Load a dataset by name and find out what is in it
- Describe a categorical variable with counts and percentages, saying which way you percentaged
- Describe a continuous variable with a centre, a spread and a statement about shape
- Quantify what missing data would cost the analysis you intend
- Look at a published statistic and ask the three questions

That's all for this week